

WINNIPEG THE FORKS, MB, Canada

WMO: 715790

Lat: 49.8885N

Lon: 97.1296W

Elev: 755

StdP: 14.30

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.6	-15.0	-26.4	1.3	-18.2	-22.6	1.6	-14.6	17.6	12.9	15.5	9.2	4.4	210	0.510

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.1	87.4	69.2	84.5	67.7	81.9	66.0	72.9	82.8	70.7	80.9	68.6	78.0	5.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.6	112.1	79.6	67.0	102.2	76.8	64.9	94.8	74.6	37.2	82.6	35.1	80.8	33.3	78.1	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.9	12.9	11.4	-23.8	93.2	4.3	2.7	-26.9	95.2	-29.4	96.8	-31.8	98.3	-34.9	100.3
			-24.1	76.8	4.2	2.8	-27.1	78.8	-29.6	80.4	-31.9	82.0	-35.0	84.0
									</					

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.4	7.0	9.1	24.4	41.3	54.2	64.8	70.7	68.3	58.9	44.1	28.2	12.1	
	DBStd	24.86	13.90	12.67	13.33	11.04	9.22	6.64	5.62	6.13	8.19	9.03	11.27	12.90	
	HDD50	5716	1334	1144	798	299	64	1	0	0	18	230	654	1174	
	HDD65	9429	1799	1564	1260	712	351	84	16	36	217	648	1103	1639	
	CDD50	2221	0	0	3	37	193	446	642	567	284	48	1	0	
	CDD65	457	0	0	0	0	15	79	192	138	32	1	0	0	
	CDH74	3678	0	0	0	6	183	621	1504	1088	259	17	0	0	
	CDH80	994	0	0	0	0	39	138	429	329	57	2	0	0	
Wind	WSAvg	5.1	5.4	5.2	5.4	5.7	5.7	4.8	4.6	4.5	4.6	5.1	5.2	5.2	
Precipitation	PrecAvg	20.6	0.7	0.6	1.0	1.4	2.4	3.3	3.0	3.1	2.0	1.4	0.9	0.8	
	PrecMax	30.6	2.0	1.9	2.7	3.9	7.1	9.1	7.6	8.7	5.9	5.1	2.4	1.7	
	PrecMin	9.3	0.1	0.0	0.0	0.0	0.3	0.2	0.8	0.2	0.4	0.1	0.0	0.1	
	PrecStd	4.4	0.5	0.5	0.6	1.0	1.8	1.9	1.8	2.0	1.2	1.1	0.6	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.9	41.1	62.6	74.0	84.8	87.9	91.5	91.5	86.0	76.1	58.5	39.7	
		MCWB	35.8	36.9	50.1	53.5	61.2	69.1	73.1	70.9	67.4	59.7	49.2	35.1	
	2%	DB	35.0	36.8	51.8	67.9	79.0	84.0	87.4	87.2	80.8	68.0	51.3	35.5	
		MCWB	32.5	33.4	44.5	51.2	58.2	65.8	70.4	69.1	65.4	55.1	44.1	32.8	
	5%	DB	31.3	32.8	45.7	63.6	74.3	80.6	84.5	83.6	76.5	62.3	47.3	32.3	
		MCWB	29.3	30.5	40.0	48.9	56.5	63.8	68.6	67.7	63.0	51.6	42.0	30.5	
	10%	DB	27.0	27.7	40.7	58.8	69.2	77.4	82.1	80.2	71.8	57.6	42.7	29.5	
		MCWB	25.2	25.6	36.6	46.7	54.3	62.7	67.1	65.7	60.8	49.6	38.3	27.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.0	37.7	53.0	56.6	65.8	73.6	76.9	75.9	71.5	60.9	50.2	35.6
			MCDB	38.9	40.8	59.4	69.4	77.1	82.4	87.7	85.0	81.6	73.1	55.8	38.8
2%		WB	32.8	33.8	44.9	52.6	61.5	69.6	73.8	71.8	67.8	56.9	45.3	33.1	
		MCDB	35.1	36.0	51.9	65.0	70.2	79.2	83.4	83.2	75.9	66.1	50.5	35.2	
5%		WB	29.3	30.2	40.1	49.9	59.2	66.7	71.5	69.7	64.9	53.0	41.7	30.5	
		MCDB	31.3	32.7	44.9	60.9	68.8	76.3	81.5	80.4	73.1	60.6	46.6	32.3	
10%		WB	25.2	25.7	36.8	47.4	56.7	64.4	69.1	67.6	62.3	49.7	38.5	27.9	
		MCDB	26.9	27.7	40.6	58.1	67.2	73.4	78.7	76.8	70.2	57.1	42.5	29.5	
Mean Daily Temperature Range	5% DB	MDBR	13.4	14.6	15.4	18.0	18.7	17.2	17.1	17.7	17.1	13.5	11.8	11.9	
		MCDBR	15.5	15.1	18.6	25.2	25.7	21.7	20.2	22.6	23.9	21.6	17.7	13.9	
		MCWBR	13.9	12.9	13.1	13.6	11.7	9.8	9.5	10.0	12.1	11.9	12.4	12.2	
	5% WB	MCDBR	15.4	14.6	17.6	22.9	19.7	18.2	18.8	19.9	19.8	19.6	16.8	13.6	
		MCWBR	13.8	12.6	12.7	13.1	10.9	10.4	10.7	10.2	12.1	11.9	12.1	12.2	
Clear-Sky Solar Irradiance	taub	0.267	0.302	0.308	0.357	0.373	0.386	0.393	0.381	0.348	0.306	0.275	0.252		
	taud	2.245	2.137	2.297	2.330	2.352	2.340	2.338	2.368	2.455	2.537	2.464	2.327		
	Ebn at Noon	246	261	285	281	281	277	273	271	270	265	248	239		
	Edh at Noon	23	33	33	36	37	38	38	35	29	22	18	18		
All-Sky Solar Radiation	RadAvg	429	816	1218	1589	1729	1850	1951	1606	1158	685	378	304		
	RadStd	46	59	101	119	143	144	81	107	90	93	53	34		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	-1.91	N/A	N/A	N/A	N/A	
(47)	Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	-1.42	N/A	N/A	N/A	N/A	

Nomenclature: See separate page